

KEN CHENG

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EDUCATION

Columbia University

Expected May 2028

B.S. in Computer Science; Statistics Minor; GPA: 3.8

Egleston Scholar (Top 1%)

- Coursework: Machine Learning, Natural Language Processing, Probability Theory, Linear Algebra, Databases

EXPERIENCE

Undergraduate Deep Learning Researcher

09/2026 – Present

Creative Machines Lab, Columbia University

PyTorch | Hugging Face Transformers | scikit-learn

- Developing LLM evaluation and activation-probing tools to study deceptive communication while distinguishing factual errors and unsupported claims
- Collected **160 Qwen2.5-7B responses** under a fixed development protocol; implemented transformer activation extraction and linear-probe tooling with scenario-level split checks and annotation gates

Software Development Engineer Intern, Amazon

06/2026 – 08/2026

Amazon SCOT, Long-Term Planning & Forecasting

Python | AWS Lambda | S3 | TypeScript CDK

- Owned design, testing, rollout, and documentation of configuration infrastructure across **17 packages and services**, now used by **30+ engineers, managers, economists, and scientists** for supply-chain forecasting
- Automated dependency tracing from research model outputs to downstream configurations, reducing manual comparison time by **~90% based on user surveys**; shipped delivery with Lambda, S3, and TypeScript CDK

Founder & Lead Engineer, Quantiv | usequantiv.com

07/2025 – Present

Deployed ML Forecasting Platform

Python | LightGBM | DuckDB | Parquet

- Trained **six LightGBM models** for earnings-move forecasting, achieving **31% lower weighted MAE** than the options-implied baseline on development validation; produced point and quantile predictions
- Automated feature and batch-inference pipelines with DuckDB/Parquet and chronological validation; published forecasts for **120+ monthly active users** with data-freshness and model drift monitoring

RESEARCH & PROJECTS

GC-INF: Graph Neural Network & Transformer Forecasting | [Paper](#)

09/2022 – 10/2023

Sole-Author Publication, IEEE ITSC 2023

PyTorch | Graph Neural Networks | Time Series

- Developed a graph-convolutional and Informer model; evaluated **four baselines across three horizons**, reducing 15-minute turning-ratio RMSE by **24%** versus STGCN on SUMO-simulated traffic data

Financial Document Retrieval Engine (FDRE) | thefdre.com

06/2026 – Present

Hybrid Retrieval, Evaluation & Citation-Verified Answers

Python | FastAPI | PostgreSQL | pgvector

- Built hybrid lexical/vector retrieval over **3.04M embedded passages from 3,204 SEC filings**, serving **75+ monthly active users** with citation checks, evidence-based abstention, and point-in-time filtering
- Raised Recall@5 from **0.152 to 0.212** with multi-query expansion on a **33-query benchmark**; cut database size **27% (15 to 11 GB)** using half-precision vectors

Wasserstein Regimes | [GitHub](#)

09/2026 – Present

Optimal Transport & Unsupervised Learning

Python | NumPy | scikit-learn

- Compared **nine clustering methods across five ETFs** using chronological folds, purged holdouts, and synthetic controls; found scale explained **95–98%** of raw Wasserstein centroid separation, limiting claims of added distributional information

AWARDS

USACO Platinum (1000/1000 Gold); **National Champion**, High School Big Data & AI Challenge (NRC)

LA Hacks 2025: Top 5 of 172 teams; Google Gemini and Financial Tech prizes (Embers)

SKILLS

Programming / Data: Python, C++, SQL, NumPy, Pandas, PostgreSQL, pgvector, DuckDB, Parquet

ML: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, LLM Evaluation, Information Retrieval

Engineering: Model Serving (FastAPI), Docker, AWS (Lambda, S3, CDK), CI/CD (GitHub Actions), Git, Linux